

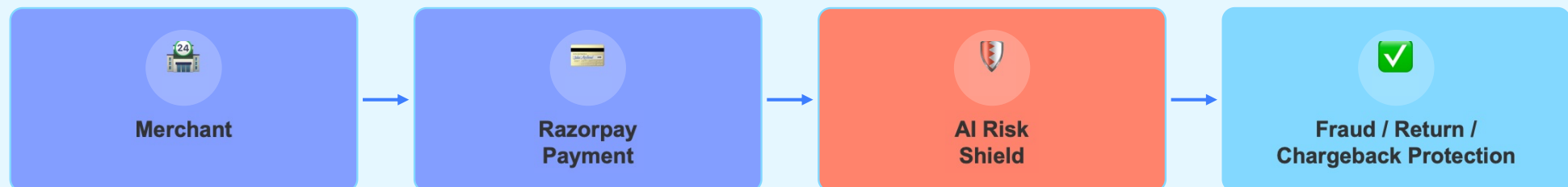
CHARGEBACKSHIELD AI

Predict • Explain • Verify • Defend

AI RISK MANAGER

Stop the merchant losing money to fraud, returns and chargebacks.

THEME	CATEGORY	TEAM ID	TEAM NAME
AI Risk Manager	FinTech / Payment Security	[INSERT TEAM ID]	[INSERT TEAM NAME]



PROPOSED SOLUTION

CORE IDEA

Converts Razorpay payment data into real-time financial risk intelligence — flagging risky transactions before they become losses.

PROBLEM ADDRESSED

- No unified view of suspicious payment behaviour
- Abnormal customer transaction patterns go unnoticed
- No early prediction of financial loss
- Little clarity on WHY a transaction is risky
- Weak, unverified evidence for chargeback disputes

SOLUTION — CHARGEBACKSHIELD AI

- Ingests Razorpay payment / webhook events
- Verifies transaction → generates Risk Score (0–100)
- Splits Fraud / Return / Chargeback risk
- Explains key drivers → estimates expected loss
- Raises alerts and prepares dispute evidence

SOLUTION FLOW

CUSTOMER → RAZORPAY PAYMENT

PAYMENT VERIFICATION → TXN DATA

AI RISK ENGINE

FRAUD
RISK

RETURN
RISK

CHARGEBACK
RISK

RISK SCORE 0–100 → EXPLAINABLE AI

ALERT

ACTION

EVIDENCE

MERCHANT DASHBOARD

Detect early → Explain clearly → Act quickly → Reduce financial exposure

UNDER THE HOOD

TECHNICAL APPROACH

TECHNOLOGY STACK

Frontend Streamlit • Plotly • HTML/CSS

Backend Python • FastAPI • REST • Webhooks

Payments Razorpay Sandbox API • Webhooks • Signature Verification

Data / ML Pandas, NumPy • Scikit-learn, XGBoost • SHAP

ALGORITHMIC APPROACH

Risk Prediction Classification model • calibration → score 0–100

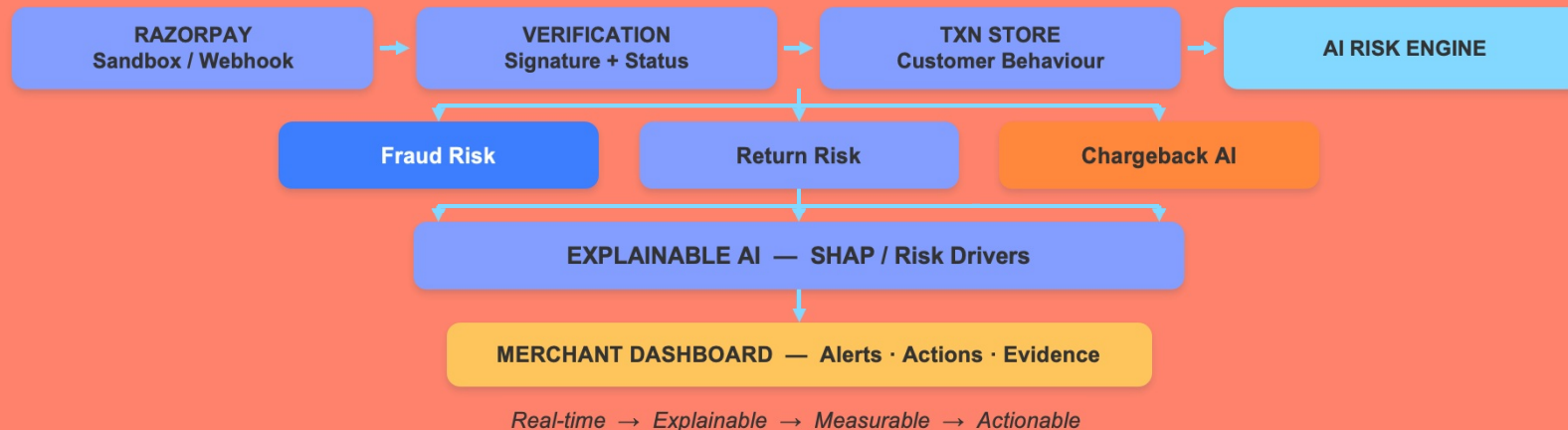
Anomaly Detection Rolling mean/SD • Z-score • EWMA

Customer Risk Frequency, spend behaviour, failed attempts, refund/chargeback history

Validation Precision, Recall, F1 • PR-AUC, ROC-AUC • FPR

$$\text{EXPECTED LOSS} = \text{Probability of Loss} \times \text{Financial Impact}$$

SYSTEM ARCHITECTURE



Real-time → Explainable → Measurable → Actionable

CAN WE BUILD IT

FEASIBILITY & VIABILITY



TECHNICAL FEASIBILITY

- Python-based modular architecture
- Razorpay Sandbox integration
- Webhook-based real-time events
- Scikit-learn / XGBoost models + Streamlit dashboard



SECURITY & RELIABILITY

- Webhook signature verification
- Env-based API secrets, input validation
- Idempotent webhook processing
- No storage of card number, CVV or OTP



KEY CHALLENGES

Challenge: False positives may block genuine customers. **Fix:** Threshold simulation + precision/recall monitoring + manual review.

Challenge: Limited historical chargeback labels. **Fix:** Use labelled data where available; separate synthetic vs real data.

Challenge: Model drift over time. **Fix:** Continuously monitor performance and risk-score distributions.



SCALABILITY

- Prototype → SME Merchant → Enterprise Risk Platform
- Future: multiple payment gateways, e-commerce platforms
- Future: ERP/CRM, bank/fintech systems
- Future: advanced fraud intelligence layer

Razorpay
Sandbox

Payment
Events

AI Risk
Engine

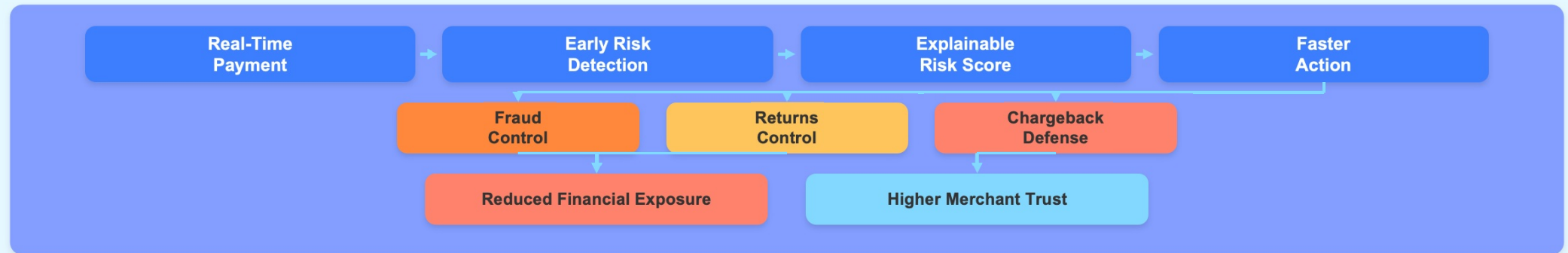
Real-Time
Decision

Merchant
Action

Loss
Prevention

WHY IT MATTERS

IMPACT & BENEFITS



FINANCIAL

- Reduce preventable fraud exposure
- Lower chargeback-related losses
- Prioritise high-value risky transactions

OPERATIONAL

- Automated risk alerts
- Faster manual review
- Centralized risk intelligence

CUSTOMER

- Fewer unnecessary payment blocks
- Better fraud protection
- Improved payment experience

BUSINESS

- Better merchant decision-making
- Improved trust
- Scalable risk management

Precision

Recall

F1 Score

PR-AUC

False Positive Rate

Expected Loss

All model-performance values must be calculated from a genuine held-out test dataset — no figures are invented.

SHOW & PROVE

PROTOTYPE & REFERENCES



Merchant Risk Dashboard

- Total transactions & average risk score
- High-risk transaction count
- Expected loss & risk trend



Live Payment Monitor

- Transaction ID, amount, customer
- Payment status
- Risk score & risk level



AI Risk Explanation

- Risk Score: 82/100 — HIGH
- Top signals: frequency, failures, amount
- Recommended action: Manual Review



Chargeback Evidence Assistant

- Payment & order verification
- Customer info & refund details
- Transaction timeline & evidence completeness

REFERENCES

- Razorpay API documentation
- Razorpay Webhook documentation
- Scikit-learn documentation
- XGBoost documentation
- SHAP documentation
- Academic research: payment fraud detection
- Academic research: anomaly detection
- Academic research: chargeback prediction

Prototype:

[INSERT WORKING DEMO URL]

GitHub:

[INSERT GITHUB URL]

DEMO MODE — SYNTHETIC DATA

RAZORPAY SANDBOX MODE